

Railroad Equipment Accidents

Data Background:

The dataset used in this assignment is on Railroad Equipment Accidents. Each row is its own accident and there are some key variables we will look at such as weather, visibility, engineering impairment, season and time of day to see if any of them correlate with railway accidents. For some such as season and time of day, there is no specific variable in the dataset for it so we will create a new one such as season and use the months to break them up into the different seasons. For time of day, we will be using variables such as AMPM and TIMEHR to find the specific time of day. The goal is to visualize and see if these correlate with railway accidents.

Visualizations:

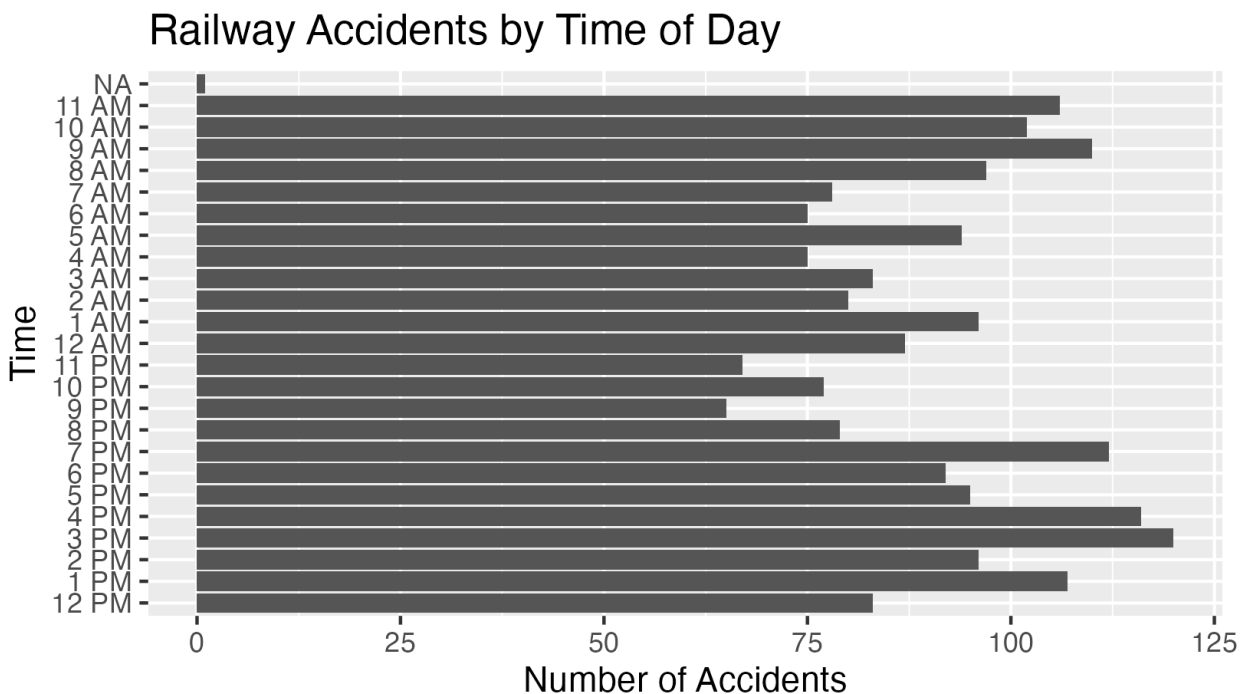


Figure 1

1. In this figure, we have on the x-axis, the number of accidents and on the y- axis the time of day the accident occurred. As we can see there is no specific time where the number of accidents occur more frequently. The highest number of accidents have occurred at 4 PM while the lowest number of accidents have occurred at 9PM. The graph shows that accidents happen throughout the day but more frequently in the afternoon and in the morning around 9 to 11 AM. There could be other factors affecting the accidents, as time of day alone is not specific enough to say.

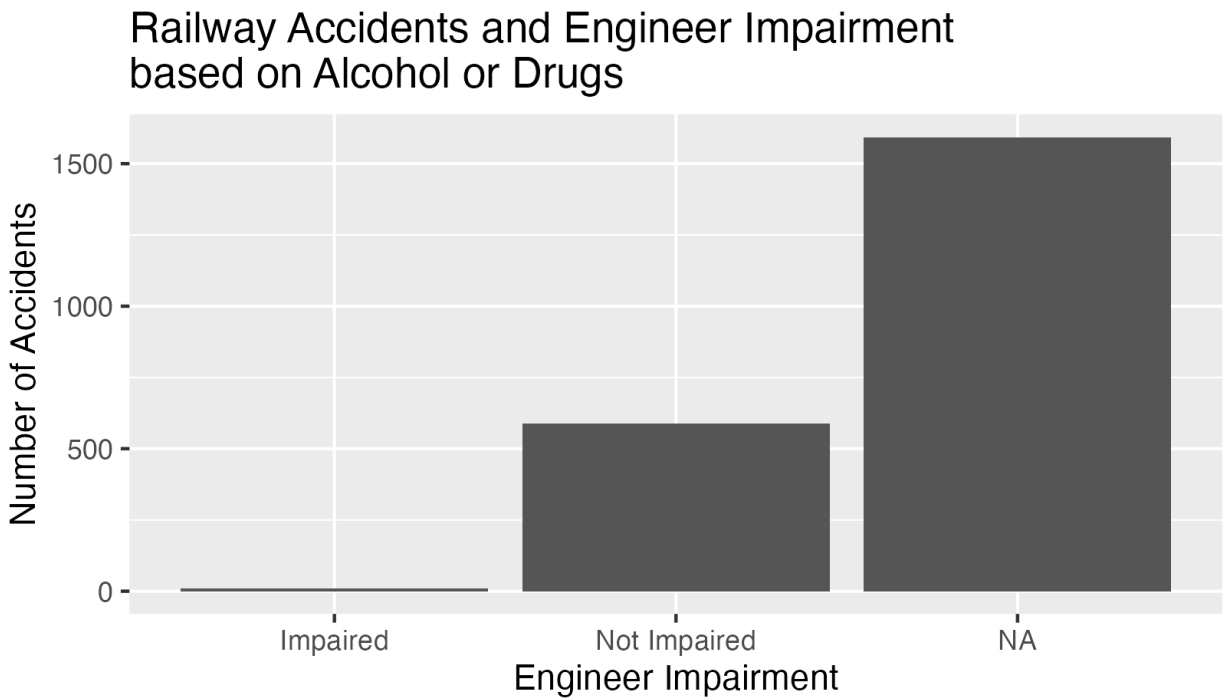


Figure 2

2. This figure shows accidents and engineer impairment through drugs or alcohol. On the x-axis we have engineer impairment and on the y-axis we have the number of accidents. We can see that the majority of accidents are not recorded for drug and alcohol use which means that there is not sufficient evidence to conclude whether or not engineer impairment is a factor in railway accidents. Of the recorded data, we can see that in over 500 accidents, the engineers were not under the influence while 11 accidents where the engineers were.

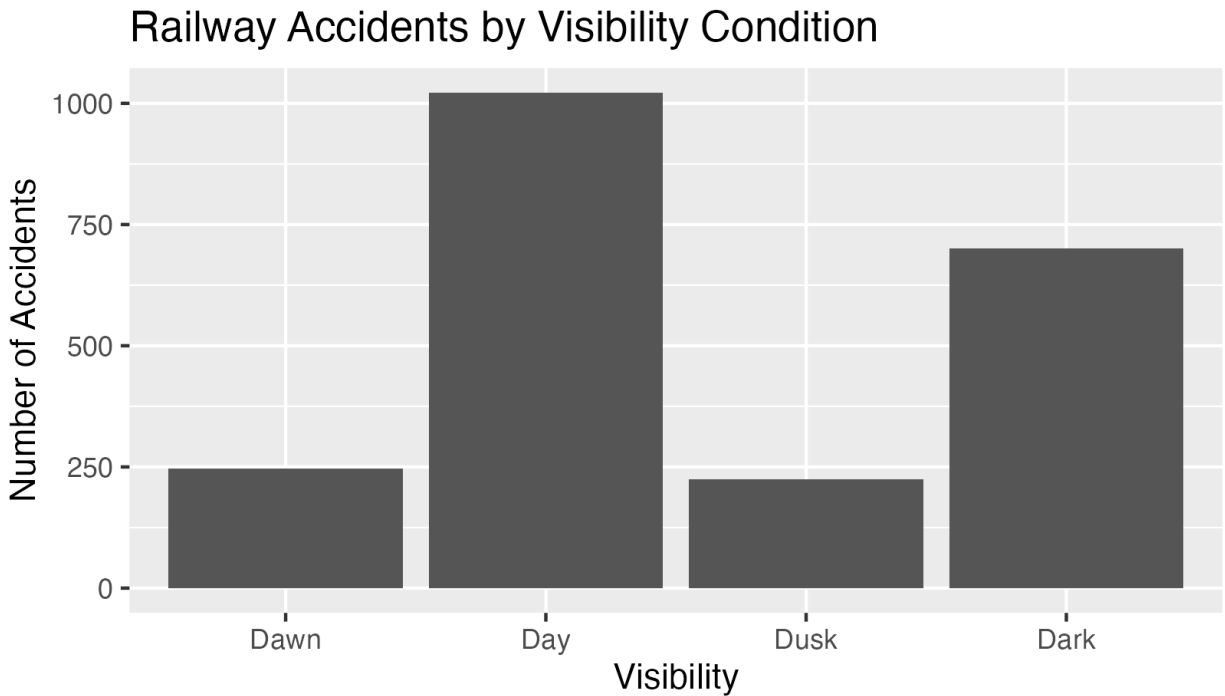


Figure 3

3. In this figure we can see railway accidents by the visibility conditions. On the x-axis we have different visibility: day, dusk, dark, and dawn. On the y-axis, we have the number of accidents. We can see that most accidents occur during the day. This could just be because more railway activity is going throughout the day. We can see the second highest is dark which could mean that low visibility is a factor in railroad accidents. Dawn and dusk are fairly equal and much less than dark and day

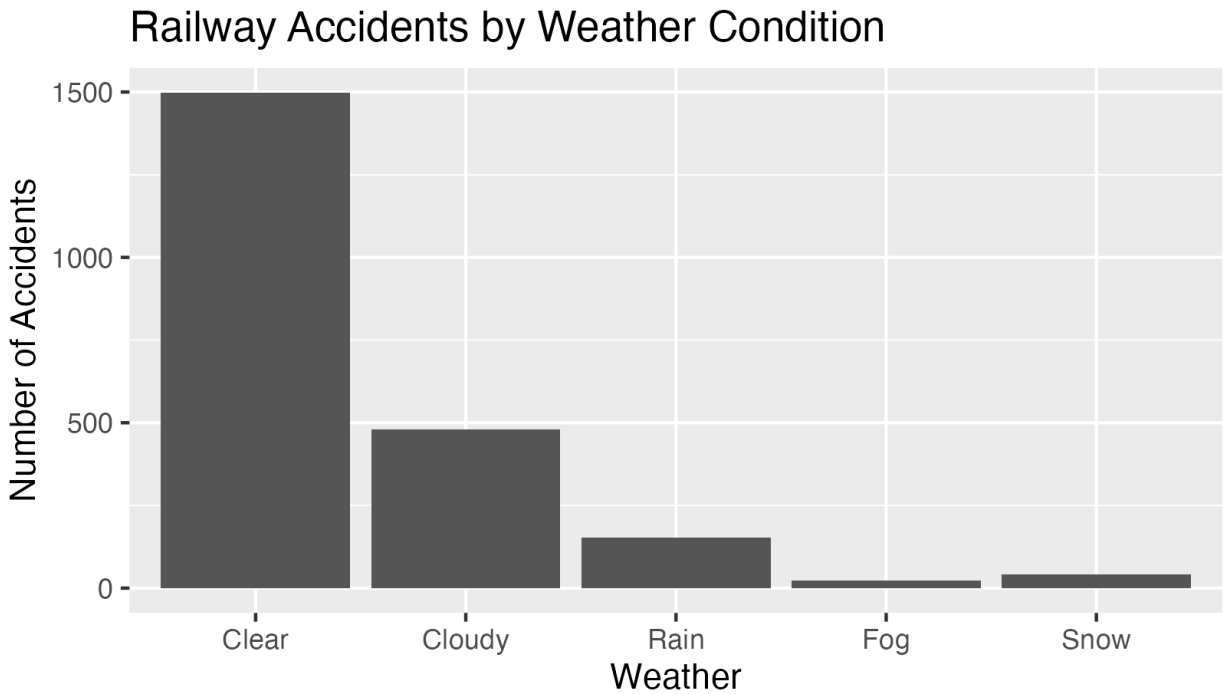


Figure 4

4. This figure shows the accidents by the weather condition. On the x-axis we have the weather conditions and on the y-axis we have the number of accidents. We can see the 5 weather conditions: clear, cloudy, rain, fog and snow. The most accidents are during clear weather conditions with the second most being cloudy. Rain makes up a good amount of accidents as well while fog and snow are small compared to the rest. Sleet is 0. We can see that most accidents happen during the day, which means poor weather conditions do not affect accidents that much. This could lead to other factors that might cause higher accidents.

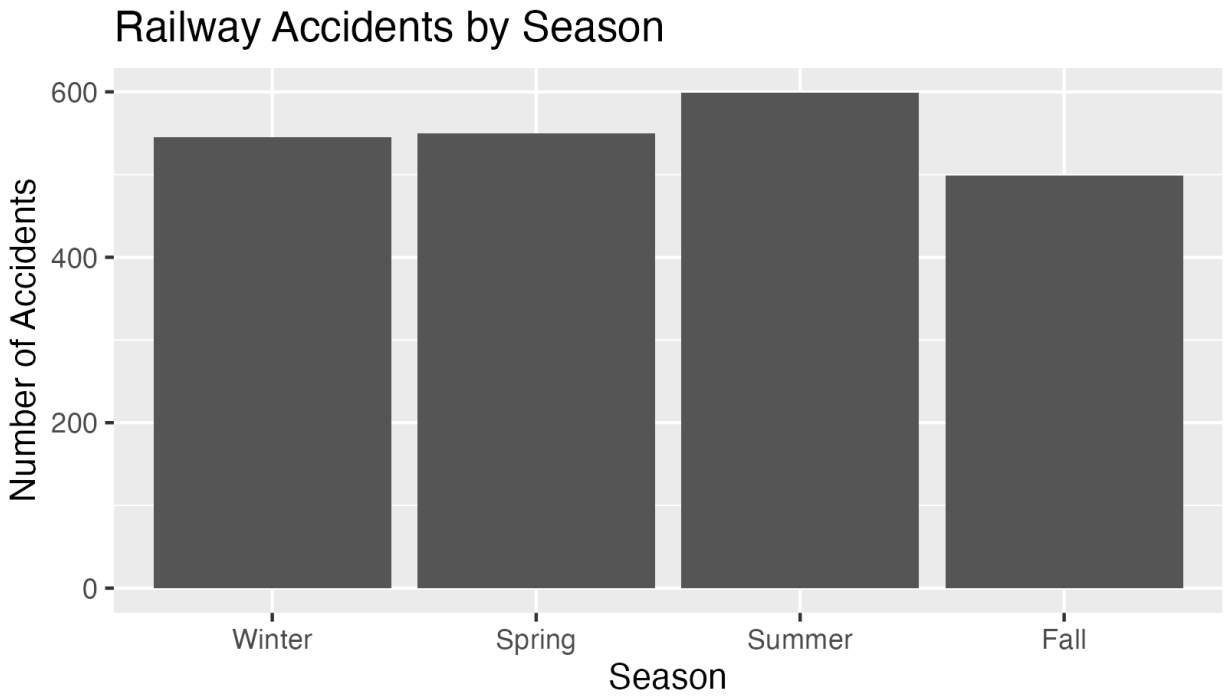


Figure 5

5. In this figure, we have railway accidents by the season. We have Winter, Spring, Summer, and Fall. On the x-axis we have the season and on the y-axis we have the number of accidents. We can see that the number of accidents is pretty evenly distributed amongst the seasons with Summer being the season with the most accidents. This shows that season does not have a direct correlation to the railway accidents as the amount of accidents in each season is around the same number.